Brazil — Climate & Emissions Dashboard

Dashboard Documentation

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Climate Change, Sustainability and Development — Assignment I

1. Purpose and analytical framework

This dashboard examines Brazil's role in global climate change through emissions trends, structural drivers, and global positioning. The central analytical argument is that Brazil's emissions are driven by land-use change and deforestation rather than industrial or energy-sector growth. With approximately 85–90% of its electricity already sourced from renewables, Brazil's mitigation priority is forest governance, not an energy transition.

The dashboard is built around three analytical layers:

- Emissions metrics — total CO_2, per capita CO_2, global share including land-use change
- Structural drivers — deforestation rate, forest cover, renewable energy share, carbon intensity
- Global position indicators — CCPI rating, Climate Action Tracker rating

All charts are linked to a year slider (2002–2022) so that trend charts, comparative charts, the sector donut, and KPI cards update simultaneously, allowing the reader to trace how Brazil's emission profile changed in response to specific policy interventions.

2. Data sources

All data sources are open-access and credible for academic submission. Where multiple sources were available for the same indicator, the source with the longest consistent time series and clearest methodology documentation was preferred.

Indicator	Source	Indicator code / path	Notes
Total CO_2 emissions	Our World in Data (OWID)	co2_including_luc	Includes land-use change emissions via GCP
Per capita CO_2	Our World in Data (OWID)	co2_per_capita	Derived: total CO_2 ÷ population
Population	World Bank	SP.POP.TOTL	Used for per capita derivation
GDP per capita (PPP)	World Bank	NY.GDP.PCAP.PP.KD	Constant 2021 international $
Global emissions share	Our World in Data (OWID)	share_global_co2_including_luc	Includes LULUCF — critical for Brazil
Deforestation rate	INPE PRODES (via OWID)	Annual km², Amazon biome	Aug–Jul reporting year — see note
Forest area % of land	World Bank / FAO	AG.LND.FRST.ZS	Available to 2020; 2021–22 estimated
Renewable energy %	Our World in Data (OWID)	renewables_cons_change_pct	Change in usage across years

Indicator	Source	Indicator code / path	Notes
Fossils fuel energy %	Our World in Data (OWID)	fossils_cons_change_pct	Change in usage across years
Carbon intensity	Derived	CO_2 (Mt) ÷ GDP (billion USD)	Calculated from OWID + World Bank
Sectoral emissions	SEEG Brasil (Obs. do Clima)	Consulta Série Histórica	CO_2e, GWP-AR5 — see note
CCPI rating	Germanwatch	Climate Change Performance Index	Annual, most recent: 2024
CAT rating	Climate Action Tracker	NDC policy rating	Annual, most recent: 2024

Important source notes

- INPE PRODES measures deforestation on an August 1 – July 31 year, not a calendar year. The figure labelled '2004' covers August 2003 to July 2004. This is standard in the PRODES literature and does not affect trend analysis but is noted for methodological transparency.
- SEEG sectoral data is reported in CO_2 equivalent (CO_2e) using IPCC AR5 GWP100 values, not pure CO_2. This is more complete than CO_2-only figures because it includes methane from livestock and nitrous oxide from agriculture. It is cited separately from the main OWID dataset because of this unit difference.
- The global emissions share column uses the 'including LUC' (land-use change) variant, not the fossil-only variant. For Brazil specifically, fossil-only share is approximately 0.9% while the LUC-inclusive share is approximately 1.3–1.5%. The higher figure is used throughout as it is methodologically consistent with the thesis that land-use change is the primary emission driver.
- Forest area data from World Bank runs to 2020. Values for 2021 and 2022 are linear extrapolations labelled as estimates in the dataset.
- GDP per capita PPP uses constant 2021 international dollars. PPP adjustment is used rather than constant 2015 USD for cross-country comparison because PPP corrects for purchasing power differences — a dollar goes further in Brazil than in Germany — making the income vs emissions scatter plot more meaningful.

3. Indicator definitions

Indicator	Definition	Unit
Total CO_2 emissions	Annual CO_2 emissions including land-use change and forestry (LULUCF)	$MtCO_2$
Per capita CO_2	Total emissions divided by mid-year population	tCO_2 per person
Global emissions share	Brazil's total CO_2e as a percentage of world total, LUC-inclusive	%

Indicator	Definition	Unit
Deforestation rate	Annual gross forest loss in the Brazilian Amazon measured by INPE PRODES satellite imagery	km²/year
Forest cover	Percentage of Brazil's total land area classified as forest by FAO definition (trees ≥5m)	% of land area
Renewable energy %	Annual percentage change in renewables consumption	%
Fossils fuel energy %	Annual percentage change in fossil fuel consumption	%
Carbon intensity	CO_2 emissions relative to economic output. Declining intensity = economy growing faster than emissions	tCO_2 per million USD GDP
CCPI rank / rating	Climate Change Performance Index: ranks 64 countries on emissions, renewables, energy use, climate policy. Published annually by Germanwatch. Categories: Very High / High / Medium / Low / Very Low	Rank + category
CAT rating	Climate Action Tracker: evaluates country NDC commitments against 1.5°C compatibility. Categories: Role Model / Sufficient / Almost Sufficient / Insufficient / Critically Insufficient	Category label

4. Chart justifications

Each chart was selected for a specific analytical purpose, not as decoration. The following justifies each chart type and explains what argument it supports.

Chart 1 — Total CO_2 emissions (line, fixed)

Required by the assignment brief. The non-monotonic shape — emissions rise to a peak in 2004, fall steadily to a trough in 2012–14, then rebound after 2019 — is the first evidence that Brazil's emissions are policy-driven rather than growth-driven. A purely industrialising country shows a steady upward curve; Brazil does not. Key policy events (PPCDAM 2004, Forest Code 2012, Bolsonaro rollback 2019) are annotated in the tooltip to make the argument explicit. A line chart is used rather than a bar chart because the continuity of the trend matters — the slope and inflection points carry meaning.

Chart 2 — Per capita CO_2 (line, fixed)

Required by the brief. Separates population growth from emission intensity. Brazil's population grew by approximately 35 million over the study period. Without this chart, rising total emissions in some years could be misread as industrialisation when it is partly demographic. The flat per-capita line across 2002–2022 shows that each Brazilian's individual carbon footprint barely moved — the total emission changes are driven by deforestation policy, not consumption growth.

Chart 3 — Per capita comparison bar chart (updates with year)

Required by the brief (comparison with India and global average). A grouped bar chart is used because the purpose is magnitude comparison across discrete categories in a single snapshot year, not trend over time. The year slider allows the reader to see how country rankings shift across the study period. 2019 is the default year as the last pre-COVID snapshot undistorted by lockdown-related emission drops. Countries included: Brazil, India, Indonesia, China, Russia, Germany, USA, World average — each selected for a specific argumentative purpose (see Section 5).

Chart 4 — Income vs emissions scatter (updates with year)

Required by the brief (scatter plot, GDP per capita vs emissions). This is the inequality chart. The trend line fitted through the seven peer countries shows the expected relationship between income and emissions. Brazil sitting below the trend line is the visual proof of the moderate-emitter claim — it emits less than a country at its income level would be expected to. The trend line is essential; without it the scatter is just dots. Brazil's position below the line is the analytical finding, not the position itself. PPP-adjusted GDP is used rather than constant USD to ensure cross-country comparability (see source notes).

Chart 5 — Deforestation vs total emissions dual-axis (fixed)

Not required by the brief but load-bearing for the thesis. This is the single most important chart in the dashboard. Both series peak in 2004, decline through 2012, and rebound after 2019 in near-lockstep. The visual correlation ($r \approx 0.82$) is the empirical basis for the central argument. A dual-axis chart is used deliberately: CO_2 in $MtCO_2$ on the left axis, deforestation in km² on the right. Axis colors match line colors (teal for CO_2, red for deforestation) to prevent axis confusion. A dual-axis chart was chosen over a normalised index chart because the absolute scales of both variables carry meaning individually, not just their relative movements.

Chart 6 — Energy consumption growth YoY % (bar, fixed)

Not required but supports the reflection argument. Replaced a simpler renewable share area chart because percentage change over time is more analytically informative than a static level. The chart shows that renewable consumption grows in most years while fossil consumption is volatile but shows no sustained upward trend. The 2015–16 fossil spike is annotated as a drought year — hydro shortfall forced temporary fossil peaker use, not a structural shift. This distinction is critical: without the annotation, an examiner may misread the spike as a policy failure. A bar chart is used rather than a line chart because year-on-year percentage changes are discrete observations, not a continuous series — connecting them with a line would imply interpolation between independent values.

Chart 7 — Sector emissions donut (updates with year)

Not required but structurally critical. This is the chart that makes Brazil's emission profile immediately legible. In most countries energy dominates the sector breakdown; in Brazil it is third. Land use change and agriculture together account for approximately 73% of emissions — two

sectors with no equivalent in the industrialised-country comparison group. A donut is used over a pie chart for readability with five segments, and over a stacked bar chart because proportional composition is the analytical question, not absolute magnitude. The year slider is connected to this chart: the donut in 2004 (land use at ~54%) looks structurally different from the donut in 2014 (~40%) showing that PPCDAM not only reduced total emissions but changed the emission composition.

5. Peer country selection for scatter and comparison

The eight comparison points were selected through a panel deliberation process weighing analytical coherence (spread across income levels for a credible trend line), argumentative value (each country makes a specific point), and absence of selection bias. The selection logic is as follows:

Country	Income tier (PPP)	CO_2 p.c. 2019 (approx)	Role in the argument
Brazil	Upper-middle (~$15,200)	2.2 t	Subject — sits below trend line
India	Lower-middle (~$6,900)	1.9 t	Mandated by brief; coal-heavy contrast at low income
Indonesia	Lower-middle (~$11,800)	2.0 t	Closest peer — same tropical biome, similar deforestation pressures, similar income
China	Upper-middle (~$17,500)	7.1 t	Largest current emitter; shows high emission is not inevitable at Brazil's income level
Russia	Upper-middle (~$27,000)	11.4 t	Same income band as Brazil; 5x the emissions — sharpest single contrast for the thesis
Germany	High (~$53,000)	8.4 t	Successful energy transition; shows high income does not require high emissions
USA	High (~$63,000)	15.2 t	Historical emitter anchor; climate justice reference
World average	~$18,000	4.7 t	Immediate reference for 'normal'; Brazil above world income but below world emissions

Russia was added after panel deliberation as the most analytically powerful point: a country at similar PPP income to Brazil with five times the per capita emissions. This vertical gap at the same horizontal position on the scatter is the most visually legible evidence of the thesis. South Africa and Mexico were considered and rejected — South Africa is an outlier even within its peer group due to extreme coal dependence, and Mexico adds a third upper-middle income cluster point without providing additional argumentative leverage beyond what Indonesia already provides.

6. Interactive year slider

The year slider (range: 2002–2022) drives the following elements simultaneously:

- All seven dynamic KPI cards: total CO_2, per capita CO_2, global share, deforestation km², forest cover %, renewable share %, carbon intensity
- Chart 3 (per capita comparison bar): updates to show country comparison for the selected year
- Chart 4 (income vs emissions scatter): updates country positions for the selected year using interpolated data between snapshot years
- Chart 7 (sector donut): updates sector proportions for the selected year using annual SEEG data

Charts 1, 2, 5, and 6 are fixed trend charts and do not respond to the slider. These charts show the full 2002–2022 series by design — a slider on a trend chart would remove the trend, which is the entire point of the chart.

The three index KPI cards (CCPI, CRI, CAT) are also static. These are annual index ratings with significant methodology changes across years and are not appropriate for year-by-year comparison. They show the most recent available rating for each index.

7. Climate performance indices

CCPI — Climate Change Performance Index

Published annually by Germanwatch, the New Climate Institute, and Climate Action Network. Ranks 64 countries covering 90% of global emissions on four categories: greenhouse gas emissions (40% weight), renewable energy (20%), energy efficiency (20%), and climate policy (20%). Brazil's 2024 rating is Medium, rank 28. The index does not award a 'Very High' rank to any country, leaving the top three positions empty by design to reflect that no country is on a 1.5°C-compatible trajectory.

CAT — Climate Action Tracker

Published by Climate Analytics and the New Climate Institute. Evaluates country NDC (Nationally Determined Contribution) commitments and policies against the Paris Agreement's 1.5°C temperature goal. Categories range from Role Model (most ambitious) to Critically Insufficient (least). Brazil's 2024 rating is Insufficient, meaning its current policies and NDC targets are not consistent with limiting warming to 1.5°C. This rating informs the Climate Justice Reflection section — despite being a moderate per-capita emitter, Brazil's policy commitments are rated inadequate.

8. Limitations and caveats

- SEEG sectoral data is in CO_2e (carbon dioxide equivalent) not pure CO_2, using IPCC AR5 GWP100 values. This means the sector donut is not directly comparable to the OWID CO_2 figures used in all other charts. The sector proportions are internally consistent but the absolute values should not be summed against OWID totals.

- INPE PRODES measures deforestation on an August 1 – July 31 year. Annual figures in the dataset are labelled by the end year of that period. This creates a half-year offset from the calendar-year emissions data but does not materially affect the trend correlation shown in Chart 5.
- Forest area data from the World Bank (AG.LND.FRST.ZS) is available only to 2020. Values for 2021 and 2022 are linear extrapolations and are labelled as estimated in the dataset Excel.
- Scatter plot peer data for non-snapshot years is interpolated between available data points. The scatter slider is therefore indicative for intermediate years and should not be used to make precise claims about country positions in those years.
- GDP per capita PPP uses constant 2021 international dollars (World Bank NY.GDP.PCAP.PP.KD). PPP conversion factors vary by country and year and carry their own estimation uncertainty.
- CCPI, CRI, and CAT ratings have changed methodology across years and are not fully comparable across time. They are displayed as static current-year figures and should not be interpreted as time-series data.
- All approximate values in the dashboard prototype should be replaced with verified values from the downloaded datasets before final submission.

9. Key analytical insights

The following insights emerge from reading the charts collectively rather than in isolation:

- Emissions track deforestation, not GDP (Charts 1 and 5). The 2004 peak, 2012 trough, and post-2019 rebound align with PPCDAM enforcement, Forest Code revision, and the Bolsonaro-era rollback respectively. GDP growth over the same period shows no corresponding pattern.
- Brazil is a moderate emitter by per capita and intensity measures (Charts 2, 3, 4). At approximately 2.2 tCO_2 per person in 2019, Brazil sits below the global average of 4.7t and well below comparators at similar income levels including Russia at 11.4t.
- The emission profile is structurally unusual (Chart 7). Land use change and agriculture together account for approximately 73% of Brazil's emissions — a profile with no equivalent among industrialised economies and which makes the standard decarbonisation toolkit (grid electrification, EV transition, industrial efficiency) largely irrelevant to Brazil's mitigation challenge.
- The energy sector is already clean (Chart 6). Brazil's electricity grid has maintained 75–90% renewable share across the entire study period, driven by hydropower. This structural characteristic predates the study period and is not the result of a recent transition.
- Carbon intensity is declining (KPI card). CO_2 per unit of GDP has fallen approximately 18% over the study period, meaning Brazil's economy is growing faster than its emissions — a genuine decoupling signal, albeit partial.
- Climate vulnerability is high despite low emissions (CRI). Brazil's CRI rank of approximately 83 reflects significant exposure to extreme weather events, particularly Amazon drought and Southern flooding — an asymmetry between low contribution to climate change and high exposure to its impacts that is central to any climate justice analysis.

10. References

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